

GARUDA

(Web application)

PROPOSAL/DPR/PROJECT PLAN

By,
Tirupati District Police

WEB APPLICATION PROPOSAL/DPR/PROJECT PLAN

1. Objective & Purpose

Clearly define:

- Why the web application is required
- Problems to be solved
- Expected outcomes
- Target users

Example:

- Crime offender database
- CCTV monitoring portal
- Pilgrim grievance system
- Cyber monitoring dashboard
- Intelligence management system

2. Scope of the Application

Define:

- What the application will do
- What modules are included
- Limitations/exclusions

Example modules:

- User login
- Dashboard
- Data entry
- Search & filters
- Reports
- Alerts
- GIS mapping
- Analytics
- Mobile integration

3. Stakeholders & Users

Identify all users:

- Administrators
- Supervisors
- Field officers
- Data entry operators
- Public users
- Technical team

Mention:

- User hierarchy
- Access permissions
- Roles & responsibilities

4. Functional Requirements

These are the actual features.

Important points:

- Login/authentication
- Role-based access control
- Data upload
- Search system
- Report generation
- Notifications/SMS/e-mail
- AI analytics
- CCTV/live feed integration
- GPS/GIS support
- Audit trail
- Mobile compatibility

Prepare:

- Module-wise functionality list
- Workflow diagrams

5. Non-Functional Requirements

Very important for government/security applications.

Include:

- Performance
- Scalability
- Reliability
- Availability
- Response time
- Security
- Backup
- Disaster recovery
- Maintainability

6. Technology Stack

Specify technologies:

a. Frontend

- HTML/CSS
- JavaScript
- React
- Angular
- Vue

b. Backend

- Node.js
- Python Django/Flask
- Java Spring Boot
- PHP Laravel
- NET

c. Database

- PostgreSQL
- MySQL

- MongoDB
- Oracle

d. Hosting

- Cloud
- NIC cloud
- State Data Center
- On-premise server

7. Database Design

Important points:

- Tables/entities
- Relationships
- Data structure
- Primary keys
- Data retention policy
- Backup strategy

Prepare:

- ER diagrams
- Data flow diagrams

8. UI/UX Design

Application should be user-friendly.

Include:

- Dashboard layout
- Navigation
- Mobile responsiveness
- Accessibility
- Multi-language support
- Dark/light themes
- Quick search options

9. Security Requirements

Very critical especially for police/government systems.

Include:

- HTTPS/SSL
- Two-factor authentication
- Encryption
- Data privacy
- IP restriction
- Role-based access
- Audit logs
- Session timeout
- Anti-hacking measures
- OWASP compliance
- Secure APIs
- Cyber security audit

For sensitive systems:

- SIEM integration
- SOC monitoring
- CERT-In compliance

10. Integration Requirements

Mention external systems to integrate:

- CCTV systems
- Mobile apps
- SMS gateway
- WhatsApp API
- Aadhaar/eKYC
- GIS maps
- Police databases
- NCRB/CCTNS
- Payment gateways
- Email services

11. Infrastructure Requirements

Specify:

- Servers
- Storage
- Bandwidth
- Backup server
- Load balancer
- Firewall
- DR site

12. Compliance & Legal Requirements

Important for official projects.

Include:

- IT Act compliance
- CERT-In guidelines
- Data retention rules
- Privacy policy
- Government cyber security standards
- Digital evidence handling standards

13. Development Methodology

Mention:

- Agile
- Waterfall
- Scrum

Development phases:

1. Requirement gathering
2. Design
3. Development
4. Testing
5. Deployment

- 6. Training
- 7. Maintenance

14. Testing Requirements

Must include:

- Unit testing
- Integration testing
- User Acceptance Testing (UAT)
- Vulnerability assessment
- Penetration testing
- Load testing
- Performance testing

15. Deployment Plan

Mention:

- Production server
- Staging server
- Rollout plan
- Migration strategy
- Backup before deployment

16. Maintenance & Support

Include:

- AMC/support period
- Bug fixing
- Future upgrades
- Helpdesk support
- SLA timelines

17. Training Requirements

For end users:

- User manuals
- SOPs
- Video tutorials
- Training sessions

18. Budget Estimation

Include:

- Development cost
- Server cost
- Licensing
- Security audit
- Cloud charges
- AMC
- Staff cost

19. Risk Assessment

Identify risks:

- Data breach
- Downtime
- Cyber attacks
- Server failure
- Insider misuse
- Lack of adoption

Mitigation measures also required.

20. Future Scalability

Mention future enhancements:

- AI integration
- Mobile app
- Facial recognition
- Predictive analytics
- Drone integration
- Multi-district rollout

21. Documentation Required

Prepare:

- DPR
- SRS (Software Requirement Specification)
- Architecture diagram
- SOP
- User manual
- API documentation
- Security policy

22. Important Diagrams

Very useful in project reports.

Include:

- System architecture diagram
- Data flow diagram
- Network diagram
- ER diagram
- Workflow diagram
- User hierarchy chart

23. Key Government/Police Application Considerations

Since many of your projects relate to police/security systems, additionally focus on:

- Real-time monitoring
- Evidence integrity
- Chain of custody
- Access logging
- Geo-tagging

- AI analytics
- Facial recognition integration
- CCTV storage policy
- Cyber forensic readiness
- Inter-departmental access control

Suggested Structure of a Web Application DPR / Proposal

1. Introduction
2. Objectives
3. Existing System Problems
4. Proposed System
5. Functional Requirements
6. Technical Architecture
7. Security Features
8. Infrastructure Requirements
9. Implementation Plan
10. Budget Estimate
11. Risk Assessment
12. Future Scope
13. Conclusion